

#Manipulating strings

1.Escape characters

```
In [6]: # \n = newline
print(" Hello!\n What is your name? \n How are you?.")
```

Hello!
What is your name?
How are you?.

```
In [10]: # \t = tab
print("Hello!\t What is your name?\tHow are you?.")
```

Hello! What is your name? How are you?.

```
In [12]: # \r = carriage return
print(" What is your name?\rHow are you?.")
```

How are you?\r name?

2.Multiline Strings

```
In [15]: print(
    """Dear Python,

    Your code has been stolen,
    and what are you do?

    Sincerely,
    Bob"""
)
```

Dear Python,

Your code has been stolen,
and what are you do?

Sincerely,
Bob

3.Indexing and Slicing strings

```
In [16]: #indexing
```

```
In [17]: str = 'Hello Python!'
str
```

```
Out[17]: 'Hello Python!'
```

```
In [18]: str[0]
```

Out[18]: 'H'

In [19]: `str[6]`

Out[19]: 'P'

In [20]: `str[-1]`

Out[20]: 'I'

In []: *#slicing*

In [22]: `str`

Out[22]: 'Hello Python!'

In [23]: `str[:]`

Out[23]: 'Hello Python!'

In [24]: `str[:5]`

Out[24]: 'Hello'

In [25]: `str[5:]`

Out[25]: ' Python!'

In [26]: `str[0:4]`

Out[26]: 'Hell'

In [27]: `str[:-1]`

Out[27]: 'Hello Python'

In [28]: `str[::-1]`

Out[28]: '!nohtyP olleH'

In [29]: `str[6:-1]`

Out[29]: 'Python'

4.'in' and 'not in' operators

in operator

In [30]: `'Your' in 'Your name'`

Out[30]: True

In [31]: `'hello' in 'hello'`

Out[31]: True

```
In [32]: 'hello' in 'Your name'
```

Out[32]: False

```
In [33]: '' in 'spam'
```

Out[33]: True

not in operator

```
In [34]: 'horse' not in 'horse and dogs'
```

Out[34]: False

```
In [35]: 'cat' not in 'horse and dogs'
```

Out[35]: True

5.upper(),lower() and title()

```
In [36]: greet = 'Welcome world!'
```

```
In [37]: greet.upper()
```

Out[37]: 'WELCOME WORLD!'

```
In [38]: greet.lower()
```

Out[38]: 'welcome world!'

```
In [39]: greet.title()
```

Out[39]: 'Welcome World!'

6.isupper() and islower() methods

```
In [41]: name = " Nareshit "
```

```
In [43]: name.islower()
```

Out[43]: False

```
In [44]: name.isupper()
```

Out[44]: False

```
In [45]: 'NARESHIT'.isupper()
```

Out[45]: True

```
In [46]: 'nareshit'.islower()
```

```
Out[46]: True
```

```
In [47]: '123'.islower()
```

```
Out[47]: False
```

```
In [48]: '123'.isupper()
```

```
Out[48]: False
```

```
In [49]: 'abc123'.islower()
```

```
Out[49]: True
```

7. is X string methods

`isalpha()` = returns `True` if the string consists only of letters.

```
In [50]: 'nareshit'.isalpha()
```

```
Out[50]: True
```

`isalnum()` = returns `True` if the string consists only of letters and numbers.

```
In [51]: 'abc123'.isalnum()
```

```
Out[51]: True
```

`isdecimal()` = returns `True` if the string consists only of numbers

```
In [52]: '123'.isdecimal()
```

```
Out[52]: True
```

`isspace()` = returns `True` if the string consists only of spaces, tabs, and new-lines.

```
In [2]: "\tHow are you?".isspace()
```

```
Out[2]: False
```

`istitle()` = returns `True` if the string consists only of words that begin with an uppercase letter followed by only lowercase characters.

```
In [5]: 'Manipulating '.istitle()
```

```
Out[5]: True
```

```
In [6]: 'manipulating '.istitle()
```

```
Out[6]: False
```

8.startswith() and endswith()

```
In [7]: 'Introduce program'.startswith('Introduce')
```

Out[7]: True

```
In [8]: 'Introduce program'.endswith('program')
```

Out[8]: True

```
In [9]: 'abc123'.startswith('abcdef')
```

Out[9]: False

```
In [10]: 'abc123'.endswith('12')
```

Out[10]: False

```
In [11]: 'Introduce program'.startswith('Introduce program')
```

Out[11]: True

```
In [12]: 'Introduce program'.endswith('Introduce program')
```

Out[12]: True

9.join() and split()

`join()` method takes all the items in an iterable, like a list, dictionary, tuple or set, and joins them into a string

```
In [14]: ''.join(['Where' , 'are' , 'you' , 'live'])
```

Out[14]: 'Whereareyoulive'

```
In [16]: ' '.join(['Where' , 'are' , 'you' , 'live'])
```

Out[16]: 'Where are you live'

```
In [17]: ', '.join(['Where' , 'are' , 'you' , 'live'])
```

Out[17]: 'Where,are,you,live'

```
In [19]: ' XYZ '.join(['Where' , 'are' , 'you' , 'live'])
```

Out[19]: 'Where XYZ are XYZ you XYZ live'

'split()' method splits a 'string' into a 'list'.

```
In [20]: 'We are live in Hyderabad'.split()
```

Out[20]: ['We', 'are', 'live', 'in', 'Hyderabad']

```
In [22]: 'Where XYZ are XYZ you XYZ live'.split('XYZ')
```

Out[22]: ['Where ', ' are ', ' you ', ' live']

```
In [23]: 'We are live in Hyderabad'.split('e')
```

Out[23]: ['W', ' ar', ' liv', ' in Hyd', 'rabad']

```
In [24]: 'We are live in Hyderabad'.split(' ')
```

```
Out[24]: ['We', 'are', 'live', 'in', 'Hyderabad']
```

10. Justifying text with `rjust()`, `ljust()` and `center()`

'`rjust`' and '`ljust`' will specify a fill character apart from a space character

```
In [26]: 'Python'.rjust(10)
```

```
Out[26]: '    Python'
```

```
In [27]: 'Python'.ljust(10)
```

```
Out[27]: 'Python    '
```

```
In [28]: 'Python'.center(10)
```

```
Out[28]: '  Python  '
```

```
In [29]: 'Python'.rjust(15)
```

```
Out[29]: '        Python'
```

```
In [30]: 'Python'.center(20)
```

```
Out[30]: '        Python        '
```

```
In [31]: 'Python'.rjust(15, '*')
```

```
Out[31]: '*****Python'
```

```
In [33]: 'Python'.ljust(10, '-')
```

```
Out[33]: 'Python----'
```

```
In [34]: 'Python'.center(10, '=')
```

```
Out[34]: '==Python=='
```

11. Removing whitespace with `strip()`, `rstrip()` and `lstrip()`

```
In [35]: str = '    Hello World    '  
str.strip()
```

```
Out[35]: 'Hello World'
```

```
In [36]: '    Hello World    '.lstrip()
```

```
Out[36]: 'Hello World      '
```

```
In [37]: '    Hello World    '.rstrip()
```

```
Out[37]: '    Hello World'
```

```
In [55]: spam = 'Where XYZ are XYZ you XYZ live'  
spam.strip('are')
```

```
Out[55]: 'Where XYZ are XYZ you XYZ liv'
```

12.The count method

```
In [39]: sentence = 'one sheep two sheep three sheep four'  
sentence.count('sheep')
```

```
Out[39]: 3
```

```
In [40]: sentence.count('e')
```

```
Out[40]: 9
```

```
In [41]: 'sentence'.count('e')
```

```
Out[41]: 3
```

```
In [42]: sentence.count('e',6)      # 6 chars since beginning of string
```

```
Out[42]: 8
```

```
In [43]: sentence.count('e', 9)
```

```
Out[43]: 6
```

```
In [44]: 'sentence'.count('e',5)
```

```
Out[44]: 1
```

13.Replace method

```
In [45]: str = 'Hello,World!'
```

```
In [46]: str.replace('World','Python')
```

```
Out[46]: 'Hello,Python!'
```

```
In [51]: fruits = ' apple,orange,mango,apple'  
fruits.replace('apple','cherry',1)
```

```
Out[51]: ' cherry,orange,mango,apple'
```

```
In [49]: fruits = ' apple,orange,mango,apple'
         fruits.replace('apple','cherry')
```

```
Out[49]: ' cherry,orange,mango,cherry'
```

```
In [52]: fruits = ' apple,orange,apple,apple'
         fruits.replace('apple','cherry',2)
```

```
Out[52]: ' cherry,orange,cherry,apple'
```

```
In [53]: sentence = "I like apples, Apples are my favorite fruit"
         sentence.replace("apples", "oranges")
```

```
Out[53]: 'I like oranges, Apples are my favorite fruit'
```

```
In [ ]:
```